



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

1. COURSE: DATA ANALYTICS AND VISUALIZATION - 24DSB3101

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TITLE OF THE PROJECT		
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Abstract

Educational institutions generate large volumes of academic, demographic, and behavioral data that can be utilized to understand student learning patterns and predict academic performance. However, conventional predictive models often function as black-box systems, providing accurate predictions without explaining the factors influencing those outcomes. This lack of interpretability limits the confidence of educators and academic administrators in adopting data-driven decision-making. To address this challenge, this project proposes an Explainable Artificial Intelligence (XAI)-based framework for Student Academic Performance Analytics by integrating comprehensive Exploratory Data Analysis (EDA), machine learning, and explainability techniques. The proposed framework utilizes three widely recognized benchmark datasets: the UCI Student Performance Dataset, xAPI Educational Dataset, and the Open University Learning Analytics Dataset (OULAD). Initially, the datasets undergo data preprocessing, including missing value handling, categorical data encoding, and normalization to improve data quality and consistency. A detailed Exploratory Data Analysis (EDA) is then performed to gain meaningful insights through dataset overview, data quality assessment, descriptive statistics, missing value analysis, duplicate record detection, outlier detection, univariate, bivariate, and multivariate analyses, correlation analysis, feature engineering, and feature selection. Following preprocessing and EDA, multiple machine learning algorithms, namely Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost, are trained and evaluated for predicting student academic performance. The performance of these models is assessed using standard evaluation metrics, including Accuracy, Precision, Recall, F1-Score, and ROC analysis. To enhance transparency and interpretability, the framework incorporates SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic

Explanations), enabling both global and local explanations of model predictions. These techniques help identify the most influential features affecting student performance, thereby making the prediction process understandable to educators and stakeholders. Furthermore, the analytical results are presented through interactive visualizations such as heatmaps, boxplots, feature importance graphs, and dashboards, facilitating effective interpretation of the data and model outcomes. Finally, a comparative analysis is conducted across all machine learning models to determine the most effective and explainable approach for student performance prediction. The proposed research framework combines educational data mining, learning analytics, machine learning, and explainable AI into a unified analytical pipeline. The project aims not only to achieve high predictive performance but also to provide transparent and interpretable insights that support evidence-based academic interventions, student success strategies, and informed educational decision-making.

Signature of the Guide

HOD